8/9/26

EXPERIMENT - 9

IMPLEMENTATION OF SUBNETTING IN CISCO PACKET TRACER

AIM:

To implement classless IP subnetting using the cisco packet tracer simulator & test connectivity between different subnets.

REQUIREMENTS:

→ cisco Packet Tracer
→ Routers
→ Switches
→ PCs
→ Ethernet cables

PROCEDURE:

1) Creating the network Topology:

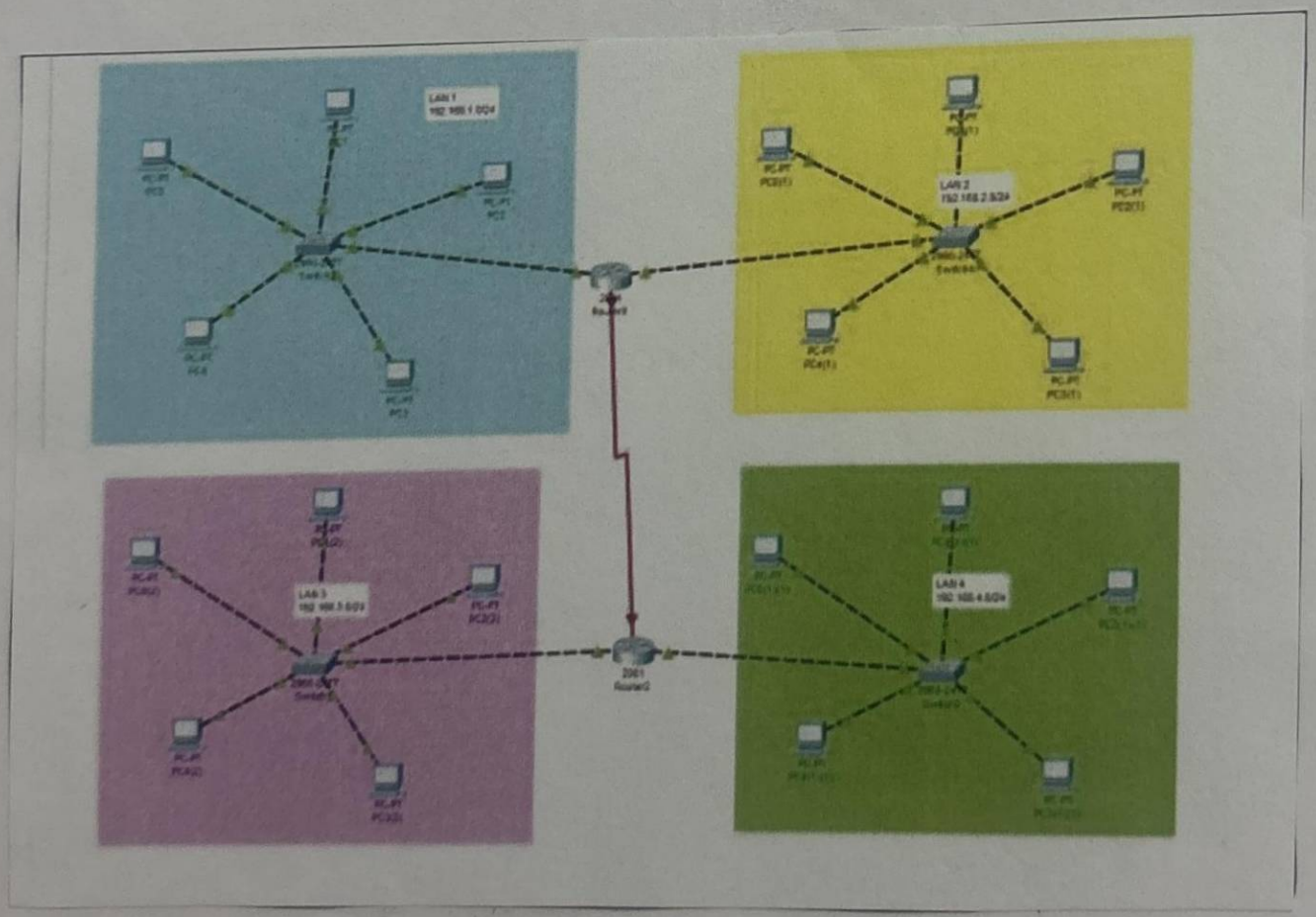
→ open cisco Packet Tracer
→ create a new workspace
→ Add the required routers, switches & PCs
→ connect the device using appropriate ethernet cable.

2) IP addressing:

LAN1 R1G0/0 192.168.1.1/27
LAN2 R1G0/01 192.168.1.2/27
LAN3 R2F0/0 192.168.1.3/27
LAN4 R2F0/1 192.168.1.4/27

For eg:

IP address: 192.168.1.1
subnet mask: 255.255.255.224
default gateway: 192.168.1.1


LAN 1
LAN 2
LAN 3
LAN 4

3) Configuring Router R1 –

Open R1 → CH & enter

enable

configure terminal

interface gigabit Ethernet 0/10

no shutdown

exit

4) Configure the ethernet:

For S1:

enable

configure terminal

interface fast ethernet 0/1

switch port mode access

exit.

5) Configure the PCs:

Open each PCs → desktop → IP configuration & enter the appropriate IP address subnet mask & default gateway.

PC	IP address	subnet mask	Default gateway
PC_1	192.168.1.11	255.255.255.244	192.168.1.1
PC_2	192.168.1.12	255.255.255.244	192.168.1.1

Testing the network:

1. Open PC → Desktop → command prompt

ping 192.168.1.1

Test connectivity between PCs in the subnet:

192.168.1.12

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num
	Successful	PC4(2)	Router2	ICMP		0.000	N	12
	Successful	PC4(2)	PC2(1)(1)	ICMP		0.000	N	13
	Successful	PC0	Router0	ICMP		0.000	N	14

RESULT:

Thus, classless IP subnetting using the cisco packnet tracer simulator & test between different subnets was successfully implemented & tested.